Chain of Command

Time limit: 3 seconds | Memory limit: 256 MB

FASTech Solutions has `N` employees sitting in a single long row, numbered `1` to `N` from left to right. Employee `i` currently has a rank a_i, and a higher number means a more senior employee.

The **boss** of employee `i` is the closest employee sitting to their left whose rank is **strictly greater** than a_i. That is, the boss is employee `j` with the largest `j < i` such that $a_j > a_i$. If there is no such employee, then `i` has no boss. Employees with equal rank are not bosses of each other.

The **chain of command** of employee `i` is the sequence formed by their boss, that boss's boss, and so on, until someone with no boss is reached. The **length** of the chain is the number of bosses in it. An employee with no boss has a chain of length `0`.

The company is busy, so ranks keep changing. You have to process `Q` events of three kinds:

- `1 l r x` — **department bonus**: the ranks of all employees `l, l+1, …, r` change by `x` ($a_i \leftarrow a_i + x$). Note that `x` may be negative.
- `2 i v` — **promotion or transfer**: the rank of employee `i` becomes exactly `v` ($a_i \leftarrow v$).
- `3 i` — **inquiry**: report the length of the chain of command of employee `i` at this moment.

Input

The first line contains two integers `N` and `Q`.

The second line contains `N` integers $a_1, a_2, \ldots, a_N$, the initial ranks.

Each of the next `Q` lines describes one event in one of the three formats above.

Output

For every event of type `3`, print the answer on its own line, in the order the events are given.

Constraints

- $1 \le N, Q \le 2 \cdot 10^5$
- $|a_i| \le 10^9$ initially
- For type 1: $1 \le l \le r \le N$ and $|x| \le 10^9$
- For type 2: $1 \le i \le N$ and $|v| \le 10^9$
- For type 3: $1 \le i \le N$
- There is at least one event of type 3.

Note that ranks can go well outside the range of a 32-bit integer after many bonuses.

Examples

Input	Output
6 8 5 3 4 4 2 6 3 5 3 6 1 2 4 3 3 5 2 4 1 3 5 1 1 6 -10 3 6	2 0 1 1 1

Explanation

The initial ranks are `5 3 4 4 2 6`.

- `3 5`: employee 5 has rank 2. Their boss is employee 4 (rank 4). The boss of employee 4 is employee 1 (rank 5): employee 3 has the same rank 4, so does not count, and employee 2 has rank 3. Employee 1 has no boss. The chain has length **2**.
- `3 6`: nobody has a rank above 6, so employee 6 has no boss: **0**.
- `1 2 4 3`: the ranks become `5 6 7 7 2 6`. Then `3 5`: the boss of employee 5 is employee 4 (rank 7), who has no boss: **1**.
- `2 4 1`: the ranks become `5 6 7 1 2 6`. Then `3 5`: employee 4 (rank 1) is not above employee 5, so the boss is employee 3 (rank 7), who has no boss: **1**.
- `1 1 6 -10`: the ranks become `-5 -4 -3 -9 -8 -4`. Then `3 6`: the boss of employee 6 (rank -4) is employee 3 (rank -3), who has no boss: **1**.